Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Diagram shows the addition of water (1) Glucose and fructose drawn with –OH groups on C1 and C2 in down positions (1)	2			2		
		(ii)		Glycosidic	1			1		
		(iii)		They have the same {chemical formula/ molecular formula/ number of atoms of each element} but different {structural formulae/ structures}/ both $C_6H_{12}O_6$ but different {structural formulae/ structures}	1			1		
		(iv)		Add Biuret solution to a sample of both solutions (1) Reject reference to heat The sucrase will cause a colour change from blue to lilac (and the sucrose will remain blue) (1) OR Heat both solutions with acid, <u>then</u> neutralise with alkali, <u>then</u> (heat) with Benedicts (1) The sucrose will cause a colour change from blue to brick red (and the sucrase will remain blue)(1)	1	1		2		2

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
	(b)	(i)		Any two (x 1) from: concentration of {sucrose / sucrase / DNS} (1) temperature (1) filter/ wavelength of light in the colorimeter (1) method of mixing (1)		2		2		2
		(ii)		Linear axes with values at origin (1) X-axis labelled pH and Y-axis labelled <u>mean</u> absorbance **+** au (1) Correct plots (1) tolerance ± ½ small square Plots joined (1) no extrapolation		4		4	4	4
		(iii)		Hypothesis is incorrect (1) The {fastest rate of reaction/ most product produced} is {pH 4/ acidic/ between pH3-5}/ the optimum pH is pH 4 (1) Use of data e.g. mean light absorption was 0.87 au at pH 4 and only 0.30 au at pH 7/ rate is approx three times faster at pH4 than at pH 7 (1)		3		3		3

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4		(iv)		3.8 (µg cm^{-3}) (1) Accept any figure between 3.7-3.8 3.8/2 = 1.9 (µg cm^{-3}) (1) OR 3.7 (µg cm^{-3}) (1) 3.7/2 = 1.85(µg cm^{-3}) (1) OR 4.2 (µg cm^{-3}) (1) Accept any figure between 4.2-4.3 4.2/2 = 2.1 (µg cm^{-3}) (1) OR 4.3 (µg cm^{-3}) (1) 4.3/2 = 2.15 (µg cm^{-3}) (1)			2	2	2	2
	(c)			There is {overlap / high variability} in the repeat readings at {pH 3/4/5/ low pH} (1) Repeat experiment with a range of intermediate values between pH 3-5/ Repeat data {to obtain a more reliable mean/ to improve reliability} (1) Reject references to accuracy			2	2		2
				Question 4 total	**5**	**10**	**4**	**19**	**6**	**15**